

AWS maintenance / establishment checklist

Station name	ZAK - AT		
People present	MCT	MARK	(Instructed by:)
Purpose of visit	FAST MAINTENANCE		
Date & time	Arrival: 30.4	14:00	Departure:
Transport	By: ARGO	From:	ZAK ON BOARD
Weather	EXCELLENT, SUNNY W/LITTL WIND		

Remarks on state of AWS / maintenance / establishment

Metadata upon arrival (Establishment of new AWS: Go to last page)

Time difference between logger clock and UTC	? +/-
Adjust logger time to UTC (be sure PC runs on UTC time)	Y/N
Download data to PC and/or <u>change CF Card</u> (wait for green light)	Y/N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y/N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Altitude (m)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north	° true/magn
Wind sensor box direction along boom? If not, measure direction.	Y/N
Radiometer aligned with mast and inclinometer? If not, measure.	Y/N
Temperature / humidity height (bottom of casing – surface)	240 cm
Sonic ranger on AWS height (membrane – surface)	112 cm
Wind sensor height (center of propeller – surface)	300 cm
Sonic ranger on stakes height (membrane – surface)	FALLEN cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	5-30 cm
Stake assembly still standing?	Y/N / more or less
Apparent damage to sensors? If yes, which?	Y/N

Remarks:

Serial numbers and maintenance details

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer			- Aligned with mast: Y/N - Clean: Y/N
Inclinometer			- Aligned with radiometer within 0.5°: Y/N
Satellite antenna			
Wind sensor			
Temperature / humidity assembly (Fan, Bracket, MP100, Hygroclip)			- New HygroClip with old assembly: Y/N - Fan spins and sounds OK: Y/N Note Ass no. + serialno. for all 4 parts !
Or: HygroClip only			- Large offset in HygroClip temperature: Y/N
Sonic ranger on AWS			- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Solar panel			- Output OK: Y/N - Clean: Y/N
GPS antenna			- Stuck to top of logger enclosure: Y/N
Data logger			- New internal battery in old logger: Y/N - New operating system: Y/N - New logger program: Y/N
Card reader			- LED either green or flashing orange (active): Y/N
Multiplexer			- Clicking as it should: Y/N
Iridium modem IMEI number			- Transmissions tested: Y/N - Problems: Y/N
Barometer			
Sonic ranger on stakes			- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Thermistor string			- New string in use, but old one still in ice: Y/N
Ablation hose assembly			- New hose in use, but old one still in ice: Y/N - Air in current hose: Y/N - Air removed from current hose: Y/N - Added 50% antifreeze mix to bladder: Y/N
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 2 / 4 / 6 / 8 / 10 m		
Tripod	- New parts: Y/N		

Metadata after maintenance / establishment

Time difference between logger clock and UTC	— + / -
Adjust logger time to UTC (be sure PC runs on UTC time)	Y / N
Download data to PC and/or change CF Card (wait for green light)	Y / N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Altitude (m)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north (radiometer should point to true south)	° true/magn
Wind sensor box direction along boom? If not, change.	Y / N
Radiometer aligned with mast and inclinometer? If not, align.	Y / N
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on AWS height (membrane – surface)	— cm
Wind sensor height (center of propeller – surface)	cm
Sonic ranger on stakes height (membrane – surface)	45+3 slow cm
Free length of stake – outer, holding boom with sonic ranger	104 cm
Free length of stake – middle	94 cm
Free length of stake – outer, holding boom without sonic ranger	114 cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm

Remarks:

28 23 32

45 3 104 94 114

52 + ? (2?)

23 + ?

28 + ?